

Literature Survey

SIH26094 | AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities

Organization	Ministry of Social Justice & Empowerment (MoSJE)
Theme	MedTech / BioTech / HealthTech
Category	Software

1. What is the Problem?

Victims and complainants under the **SC/ST (Prevention of Atrocities) Act, 1989** go through a long, stressful legal journey — not a one-time event.

- After filing a complaint, victims face **threats, intimidation, repeated court appearances, investigation/trial delays, social boycott, and financial hardship**.
- Current government mechanisms (helplines, portals) focus on **legal and financial support** — registering the FIR, tracking the chargesheet, releasing compensation.
- **Nobody is continuously watching how the victim is coping mentally** while all this is happening. Distress usually becomes visible only after a crisis (victim withdraws the case, self-harm, breakdown in court, etc.) — too late for timely help.

2. What Solution is Needed?

An **AI system that checks in with the victim regularly** (not just once) across the entire investigation → trial → rehabilitation → compensation journey, and warns officials **before** a crisis happens.

- **Regular check-ins** via chatbot, IVRS call, SMS, mobile app or the NHAA helpline itself.
- **NLP + Sentiment Analysis + Emotion AI + Voice Stress Analysis** on what the victim says/types/sounds like.
- A **Dynamic Distress Score** that is tracked as a trend over weeks/months — not a single snapshot test.
- **Automatic alerts** to counsellors, district officers and designated officials when the score crosses a risk threshold.
- **Actionable recommendations**: counselling, medical help, witness protection, relocation, financial aid, legal aid.
- **Dashboards** at district / state / national level, built with **explainable AI, data privacy and security** at the core.

3. What Already Exists?

Several pieces of this puzzle already exist in India and globally — but as **separate, disconnected systems**.

- **Tele-MANAS (14416)** – Ministry of Health's national tele-mental-health helpline. Offers 24/7 counselling, psychiatric consultation and an AI chatbot for triage, in 20+ languages. Has handled **3.4+ million calls** since 2022.
- **NHAA / SAMBAL (14566)** – MoSJE's own helpline and web portal for atrocity victims. Tracks **FIR registration, investigation and chargesheet status**, and lets victims/NGOs check grievance status online.
- **Witness Protection Scheme, 2018** – a Supreme-Court-approved framework giving victims/witnesses categorized physical protection (police escort, identity change, relocation) for a fixed period.

- **Global AI crisis-detection tools** – e.g. Crisis Text Line's machine-learning model, which identifies roughly **85–90% of texters at severe/imminent risk** from message text alone; consumer chatbots like **Wysa and Woebot** use emotion detection to hold supportive conversations.
- **Academic prototypes (2025)** combining facial emotion recognition, speech emotion recognition and chat/scale-based assessment into one “holistic” monitoring pipeline.

4. What is Lacking in These Solutions?

None of the above was built **specifically for atrocity victims tied to their legal case timeline** — this is the real gap SIH26094 needs to close.

- **Tele-MANAS is general-purpose**: it doesn't know a caller is a PoA Act victim, doesn't track the same person over months, and recent reports show a rising share of AI conversations still need to be handed off to a human because the AI under-performs on nuanced distress.
- **NHAA/SAMBAL tracks the case, not the person** — there is no mental-health score, no trend line, and no alert if a victim is quietly deteriorating between hearings.
- **Witness Protection Scheme, 2018 is only a physical-security guideline**, not backed by law: **weak implementation, uneven rollout across states, limited funding**, and protection orders lapse after 3 months and need re-application — there is no continuous digital well-being layer at all.
- **Global crisis-AI tools have known limits**: false positives on ordinary language (e.g. flagging “not suicidal, just depressed” as high-risk), and **bias against non-English / regional-language speakers** since most models are trained on Western, English-language data — not tuned for India's languages or the caste-violence context.
- **No system links case events to distress**: a delayed chargesheet, an upcoming hearing, or the accused getting bail are exactly the moments distress is likely to spike — yet no existing tool connects legal-case data with mental-health monitoring.

5. Which Department Needs This, and How to Add Value

Owner: Ministry of Social Justice & Empowerment (Department of Social Justice and Empowerment) — it already runs NHAA/SAMBAL, the Integrated Portal, and PoA Act victim relief/compensation, so this system extends its existing mandate rather than starting fresh.

To make the solution genuinely valuable (not “just another chatbot”):

- **Plug into NHAA/SAMBAL and the Integrated Portal** instead of building a new parallel database — reuse the same case ID so distress data is linked to real case milestones.
- **Reuse Tele-MANAS's counsellor network** for escalation/hand-off instead of building a separate call centre from scratch.
- **Support regional languages and voice heavily** — many victims are rural or semi-literate, so text-only chat will exclude them.
- **Build in Explainable AI and DPDP Act, 2023-compliant privacy** from day one — this is sensitive caste-violence + mental-health data; a leak could endanger the victim.
- **Add case-timeline-aware prediction** — correlate hearing dates, bail status and chargesheet delays with the distress score. This is the genuinely new piece no existing system (Tele-MANAS, SAMBAL, or global tools) currently does.
- **District/state dashboards for SC/ST welfare officers** to prioritise the highest-risk victims first, closing the loop between detection and real-world action.

Sources

- PIB – 14566: National Helpline Against Atrocities on SCs/STs — pib.gov.in

- National Helpline Against Atrocities (NHAA / SAMBAL) portal — nhapoa.dosje.gov.in
- Tele-MANAS official portal, Ministry of Health & Family Welfare — telemanas.mohfw.gov.in
- India's Tele-MANAS: evolution, early outcomes — British Journal of Psychiatry / Cambridge Core
- Tele-MANAS callers turning to humans after AI falls short — Vision IAS / The Hindu, Apr 2026
- Speech Emotion Recognition in Mental Health: Systematic Review — JMIR Mental Health, 2025
- AI-Powered Holistic Mental Health Monitoring (FER + Chatbot + Voicebot) — ResearchGate, 2025
- Crisis Text Line – Detecting Crisis: An AI Solution — crisistextline.org
- Crisis Text Line case study — bestpractice.ai
- Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Act, 1989 — ncsk.nic.in
- Witness Protection Scheme, 2018 – Challenges & Reforms — International Journal of Law Management & Humanities (ijlmh.com)
- Witness Protection Scheme: Why India needs it — ClearIAS